Type MS · Calculate an amount from a percentage, fraction or probability

Answer 1 of 14 · Question 7 part c.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part c.i in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 1 of 14 · Question 7 part c.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 1 of 14 · Question 7 part c.i · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 2 of 14 · Question 4 part a · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	$\frac{30\,000}{100}$ x 40; 12 000;	Full marks for final correct answer ecf	1 1
(b)	a microorganism/named microorganism that causes disease;		1
(c)	It cannot be transmitted from one person to the next		1
(d)	• use a sample of people/use two groups of people; • feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B; • monitor health/compare (health) of two groups;		1 1 1
(e)	• breakdown of cartilage/cartilage not formed properly/damaged; • bones rub together/more friction between bones/less shock absorption;		1 1

Total question 4 = 9 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 2 of 14 · Question 4 part a · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 3 of 14 · Question 4 part c.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

number				
4 (a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
	(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
	(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
	(ii)	0.9 x 3.8 = 3.4(2) million;		1
(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 3 of 14 · Question 4 part c.ii · 1 mark

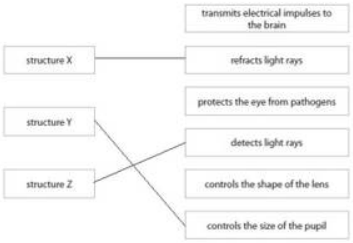
2021 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 4 of 14 · Question 1 part b.iii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

number					
1	(a)	(i)			3
		(ii)	A; B, C and D do not describe stereoscopic vision.		1
	(b)	(i)	build-up of protein/cloudy/opaque; in the lens;		2
		(ii)	surgery/operation to replace the lens/lens replacement;		1
		(iii)	39 000^{0}000 x 43 ÷ 100; 16.77 million/16.8 million/16 770 000;		2
		(iv)	vitamin A;		1

Total 10 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 4 of 14 · Question 1 part b.iii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 5 of 14 · Question 5 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
5	(a)	• change in base sequence; • causes change in protein synthesis; • causes change in phenotype;		3
	(b) (i)	• axes labelled with units; • suitable scale; • correct plots; (3 x 2) • bars of equal width;		5
	(ii)	• 43 000 × 0.0015; • = 64.5; • 65;		3
	(c)	• extra chromosome at pair 21; • diploid number 47; • female because 2 X chromosomes; • at pair 23;		4

Total : 15 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 5 of 14 · Question 5 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 6 of 14 · Question 6 part a.i · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

6 (a) (i)	2500 males; $\frac{2500 \times 0.8}{100}$; = 20;		ecf if use 5000 as starting figure 40 scores 2 marks any number×0.8=1	3
(ii)	small number/ not enough tested;			1
(iii)	• many more males than females with condition; • show that it is sex-linked; • to X chromosome; • females are XX /males one X chromosome/XY • females can be carriers(so fewer)/males cannot be carriers(so more); • (relatively) small number (in whole population)/small chance of being colour blind; • indicates that it is recessive;			5
(b)	parents	$X^BY \times X^BX^b$;		4
	gametes	X^B Y X^B X^b;		
	offspring genotypes	$X^B X^B$ X^BX^b X^BY X^bY;		
	offspring phenotypes	normal carrier/ disease female normal female male red-green male;	If parents are incorrect ecf for mp 2,3 and 4 If X and Y are not used- max of 1 mark	
				Total = 13

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 6 of 14 · Question 6 part a.i · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 7 of 14 · Question 4 part b.ii · 2 marks

2024 JUN · Paper 02 · find part b.ii in this mark scheme extract

Number				
4 (a)	(i)	Correct labels to: trachea; bronchus/bronchi; alveoli;		(3)
	(ii)	• many alveoli/large surface area; • short diffusion distance/ alveoli 1 cell thick/capillary 1 cell thick/ thin wall; • moist; • (dense/extensive/large) network of capillaries;		(4)
(b)	(i)	any four from • reference to use of a nose clip; • reference to soda lime (to absorb carbon dioxide) • reference to sterilising mouth piece; • breathe in and out(normally) through a mouthpiece(connected to spirometer); • pen goes up and down (on a rotating drum);\ • measure peak to trough/ highest to lowest/top to bottom/ from the trace;		(4)
	(ii)	4600 ÷ 30 (x 100) = 1380; 4600 – 1380 = 3220 (cm^3); or 4600×0.7=3220 (cm^3);;	ecf: allow correct calculation from values given for mp1. Max 2 marks	(2)

Total for Question 4 = 13 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 8 of 14 · Question 5 part b.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

Number				
5	(a) (i)	Any two from: • chemical; • diffused from one neurone to another; • allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones;		2
	(ii)	• neurotransmitter/dopamine/ diffuses/travels across synapse; • binds with receptor (on postsynaptic membrane)		2
	(iii)	Any two from: • increased movement; • increased heart rate/blood pressure; • more energy; • sleeping difficulties; • anxiety/stress; • hallucinations;		2
	(b) (i)	Any two from • tremors/random muscular movement / involuntary movement; • slower movement; • speech/writing changes; • impaired posture/balance; • sleeping problems; • rigidity/stiffness; • anxiety/depression/mental health/memory problems;		2
	(ii)	• (145 000 ÷ 100) (x 18) = 26 100; • (26 100 + 145 000) = 171 100;	Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark Full marks for correct final answer	2

Total for question 5 = 10 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 9 of 14 · Question 4 part a.iii · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

4 (a) (i)		<table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table> • two row/column table; • concentration (sugar) in au; • percentage/% change in mass; • 0.0 to 1.50/1.50-0.00;	Ignore units in the body of the table Accept AU for MP2 Ignore % symbol for MP3 mp3 unit not required for percentage in mass	4
	(ii)	0.50 (au);		1
	(iii)	$\frac{104 \times 7}{100}$; = 7.28 g; =7.3;	Allow 4/100 for 1 mark or 0.28	3
(b)		• change in mass is -1% to +4% • 5% range in mass • between 0.25 and 0.5 au; • accept any answer between 0.375- 0.5; ***an answer of o.45 scores =4 marks***	MP 1-4 are separate MPs unless the answer is 0.45	4
(c)	(i)	• dry the cylinder/ remove water/liquid; • affects mass/wrong mass/heavier/inaccurate mass/eq;	Accept weight	2
	(ii)	any two from • volume of (sugar) solution; • temperature; • surface area of cylinders; • same type of potato;	Ignore amount accept volume of liquid	2 **Total =16**

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 9 of 14 · Question 4 part a.iii · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 10 of 14 · Question 2 part c · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Number				
2 (a)	(i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
(b)		D; (medulla oblongata)		1
(c)		$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
(d)		• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 10 of 14 · Question 2 part c · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 11 of 14 · Question 7 part a · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7 (a)	$\frac{14}{100}$ x 250 = 35;	Ecf one mark for correct answer from wrong numbers	1
	35 ÷ 10 = 3.5	Full marks for correct final answer of 3.5	1
(b) (i)	4 x 3.5 = 14 (units);	ecf from 7a	1
	14 – 3 = 11 (hours);	Full marks for correct final answer of 11 hours	1
(ii)	Any two from: • alcohol is a depressant; • (alcohol) affects the nervous system/brain; • slows reactions/increases reaction time;		Max 2
(c)	Any five from: • less/no bile produced; • less/no emulsification of fat; • surface area of fat remains low; • less fat digested/broken down (by lipase); • pH of small intestine remains low/acidic; • enzymes denatured/reduced enzyme activity; • reduced/no digestion by protease / carbohydrase/fewer proteins/carbohydrates digested/broken down; • more fat egested;	Allow trypsin for protease and amylase for carbohydrase	Max 5

Total question 7 = 11 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 11 of 14 · Question 7 part a · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

7 (a)	Any four from: • less blood flow/blood flow slows/restricted; • to heart muscle/tissue/cells; • less oxygen/glucose; • less aerobic respiration; • heart muscle/tissue/cells die/angina/heart attack;		(4)
(b) (i)	• (women 29 ÷ 100 (x 33 500 000) 9 715 000; • (men 26 ÷ 100 (x 33 000 000) 8 580 000; • (9 715 000 – 8 580 000) 1 135 000/1.135 million/1.135×10^6;	ecf for mp3	(3)
(ii)	• calculate BMI; • mass÷(height)^2; • compare result to BMI chart/ obese if 30 or more;	Accept • waist: hip ratio; • waist÷hip • Compare result to hip:waist ratio chart/	(3)
(c)	• beta blockers bind/ attach to receptors (on cardiac muscle); • prevent neurotransmitters from binding/attaching (to receptors); • reduce heart rate/reduced blood pressure;	Accept beta blockers fits into receptors accept neurotransmitters do not fit into receptors	(3)

Total for Question 7 = 13 marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 13 of 14 · Question 3 part b.i · 3 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 13 of 14 · Question 3 part b.i · 3 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 14 of 14 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Calculate an amount from a percentage, fraction or probability

Answer 14 of 14 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks